

H1 — ML + Embedded Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the C runtime matches PyTorch on this machine.

Steps: 1. Set up the Python env with `uv sync`. 2. Build the portable C runtime on host. See runtime/host_verify. 3. Run the host golden tests: tests/ and runtime/host_verify. 4. Record the max abs diff.

Deliverable: the golden test result in the tracking sheet.

Verify: the diff is 1e-5 or below, the published value.

Learning this week

What a model does at runtime: token in, logits out. Read the first 100 lines of the runtime main loop. Write what each of these does in one line: embeddings, attention, FFN, output head.

Resources

- runtime/: <https://github.com/slvDev/esp32-ai/tree/main/runtime>
- tests/: <https://github.com/slvDev/esp32-ai/tree/main/tests>
- RESULTS.md (golden claim): <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>

Done when

- [] Golden diff is 1e-5 or below
- [] Result is in the tracking sheet
- [] You can say what the project does in one sentence